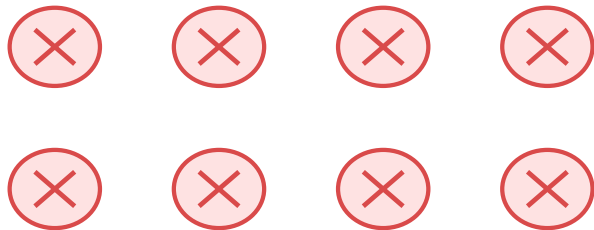


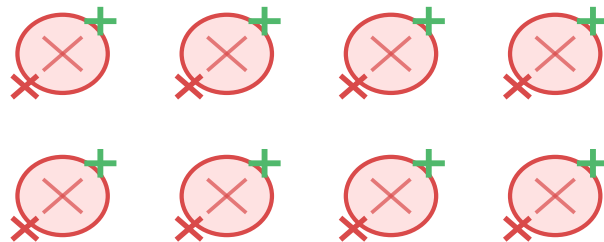
Outcome-only GRPO



all-fail group $\rightarrow$ advantage = 0

NO GRADIENT (dead update)

RLVP



+ discharge credit **X** step penalty

process rewards $\rightarrow$ per-step credit

GRADIENT from failed episodes